

SHETH L.U.J AND SIR M.V COLLEGE

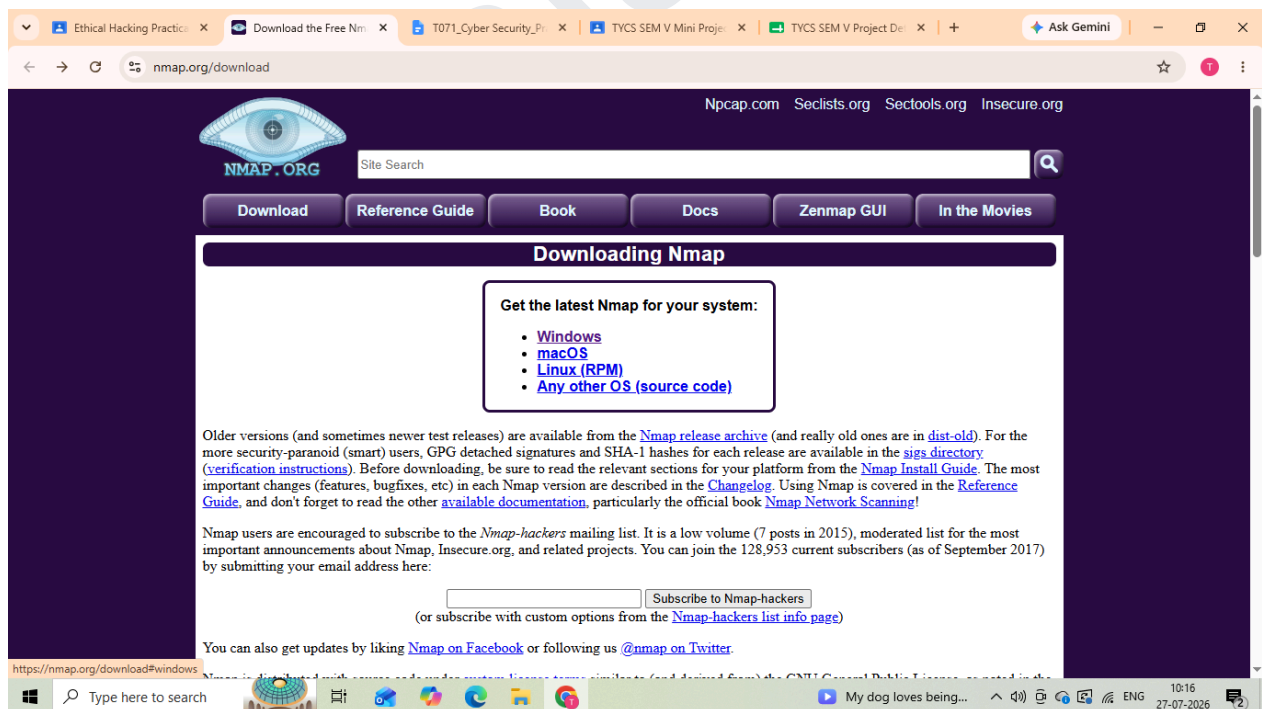
SUBJECT:- ETHICAL HACKING

Practical no: 4

Aim:Port Scanning with NMap

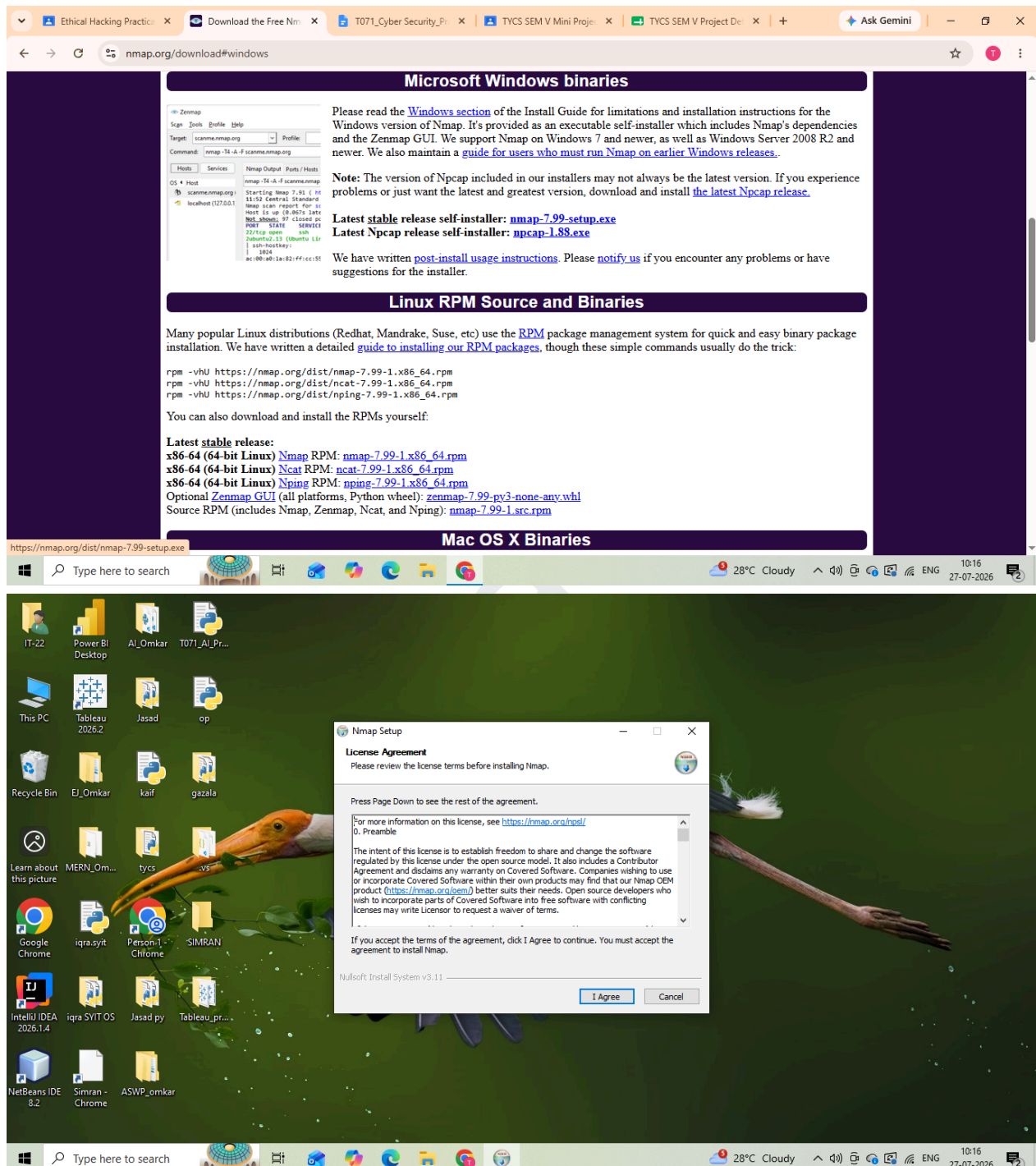
- Use NMap to perform an ACK scan to determine if a port is filtered, unfiltered, or open.
- Perform SYN, FIN, NULL, and XMAS scans to identify open ports and their characteristics.
- Analyze the scan results to gather information about the target system's network services.
- Use NMap to perform an ACK scan to determine if a port is filtered, unfiltered, or open.

Step 1 : Install nmap from : <https://nmap.org/download> and open it.



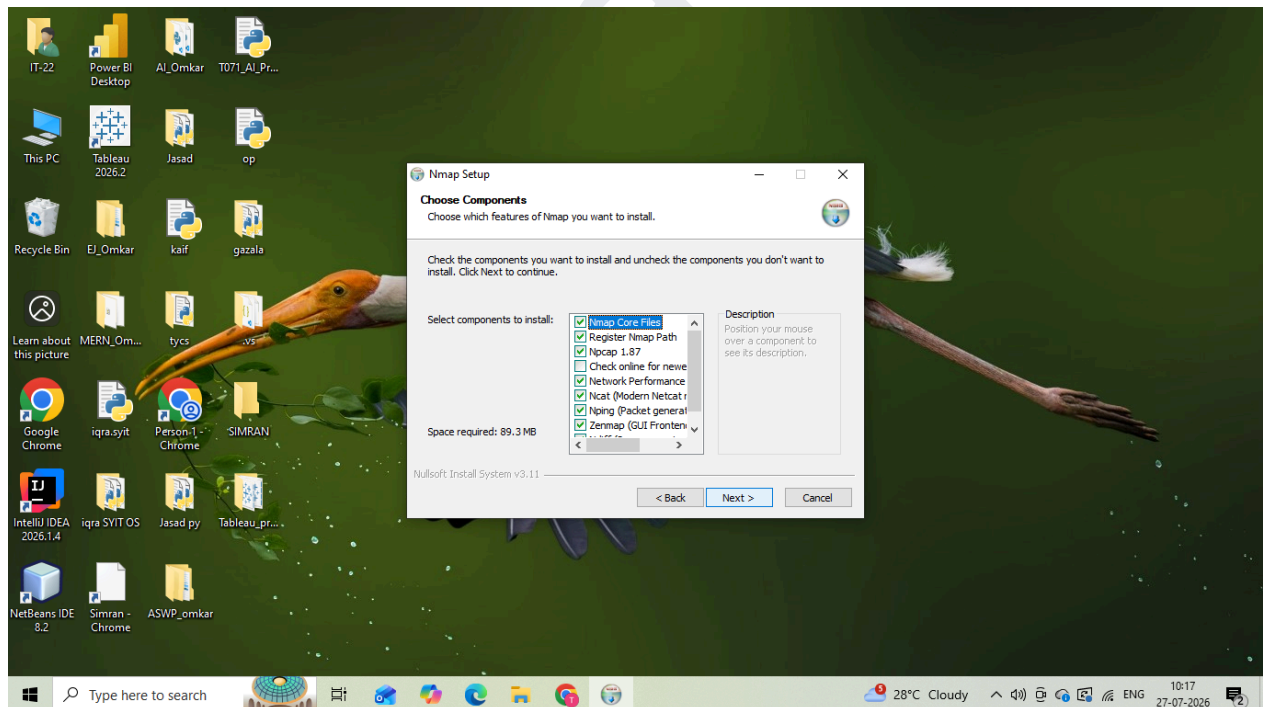
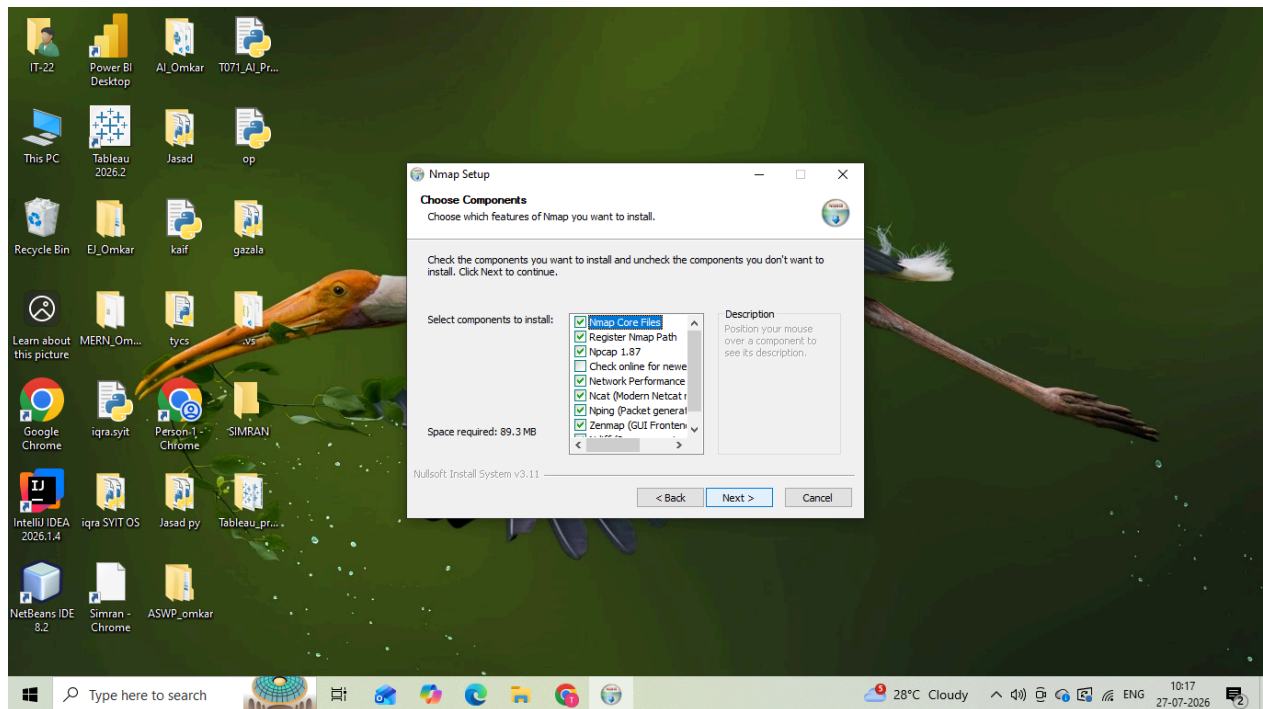
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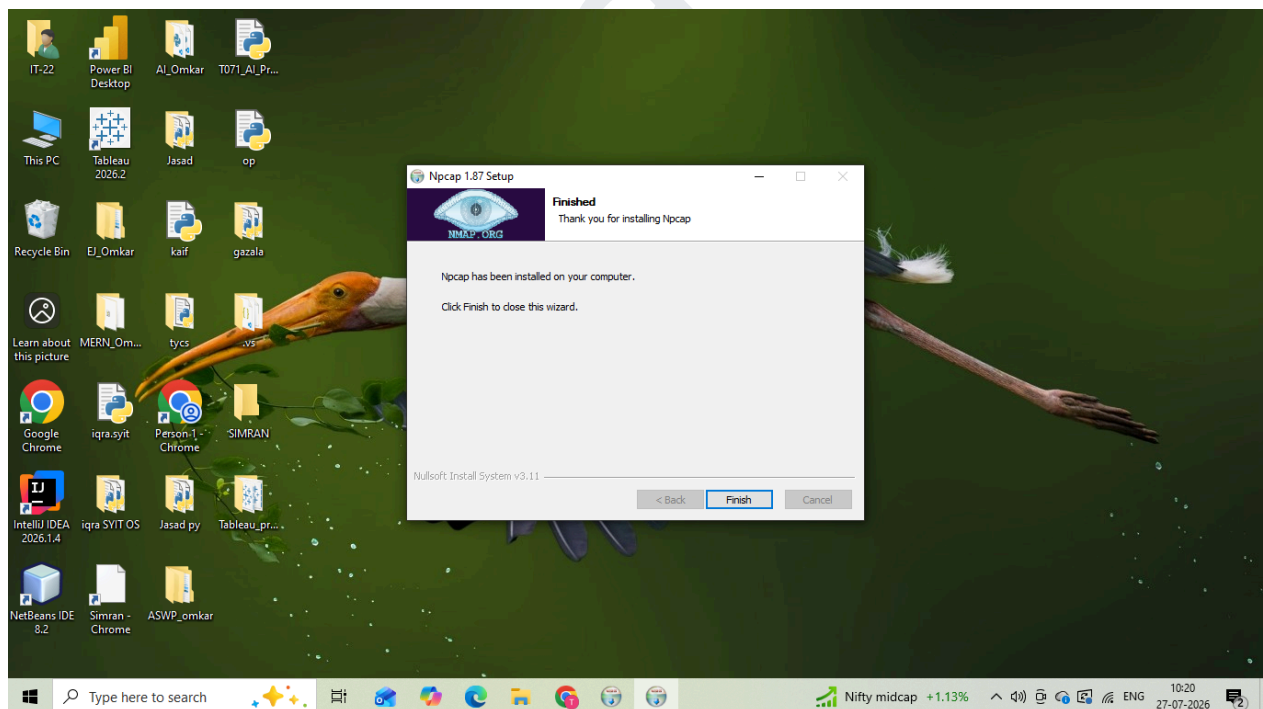
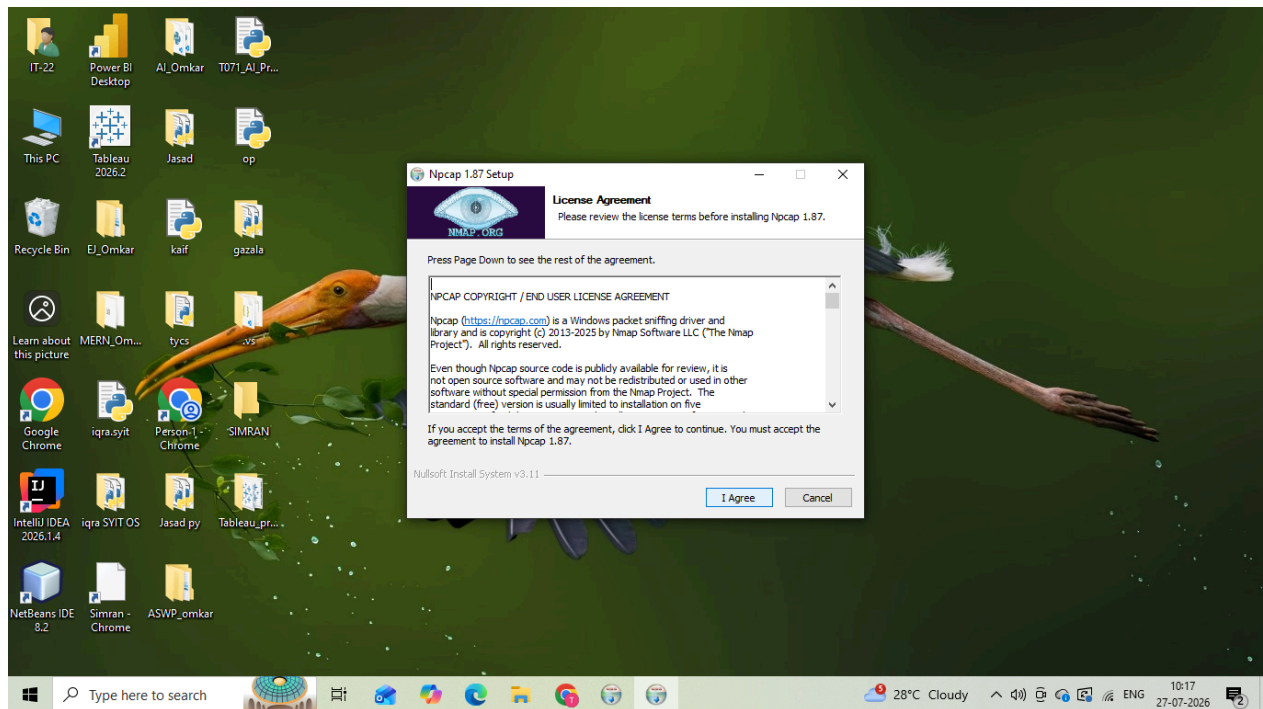
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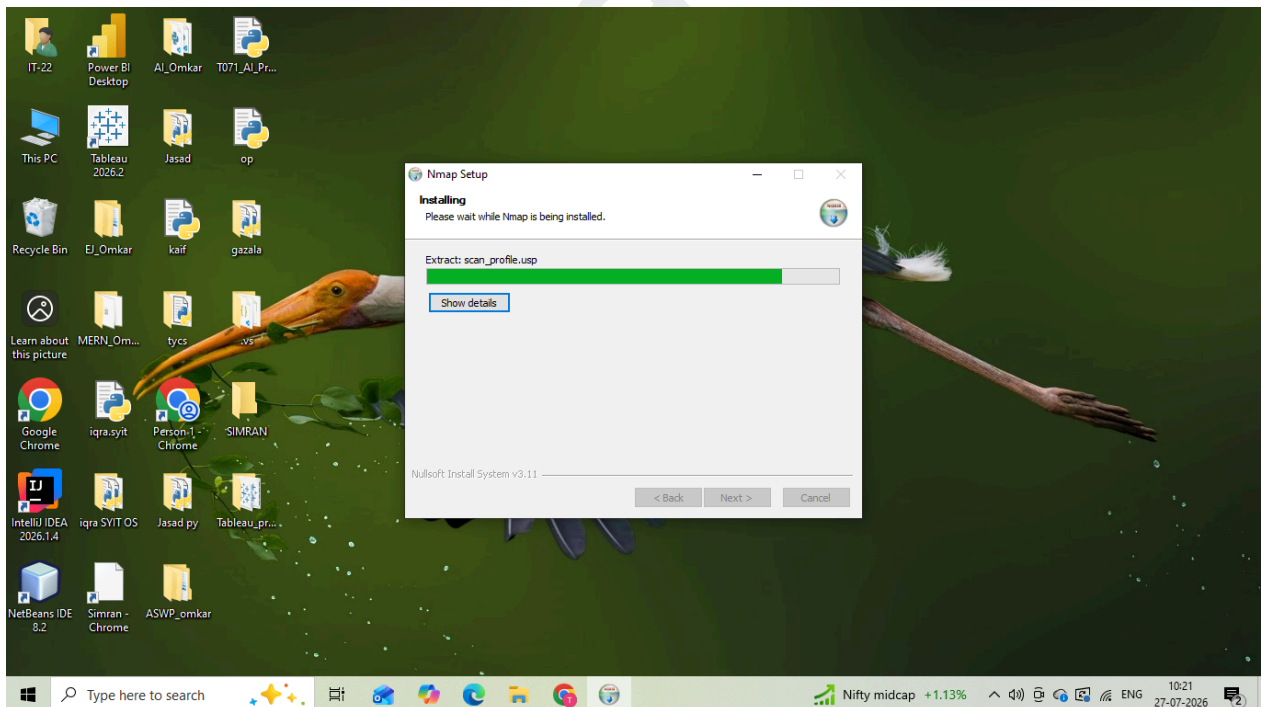
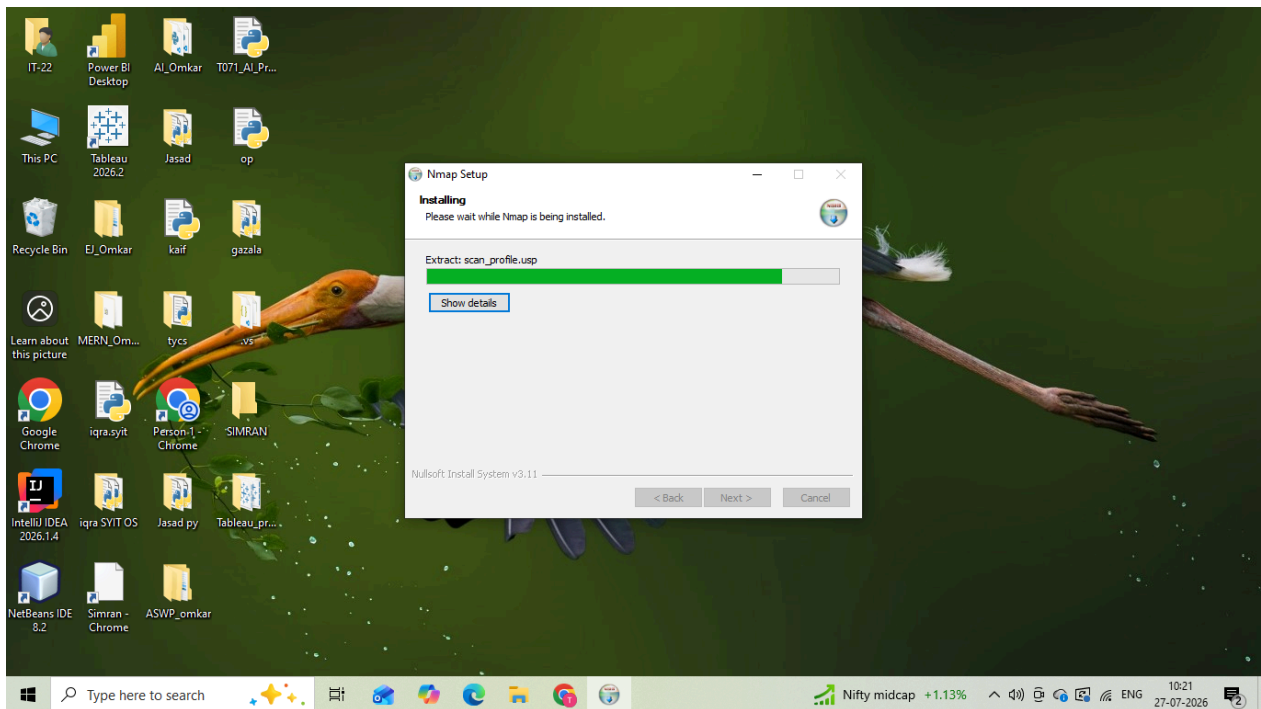
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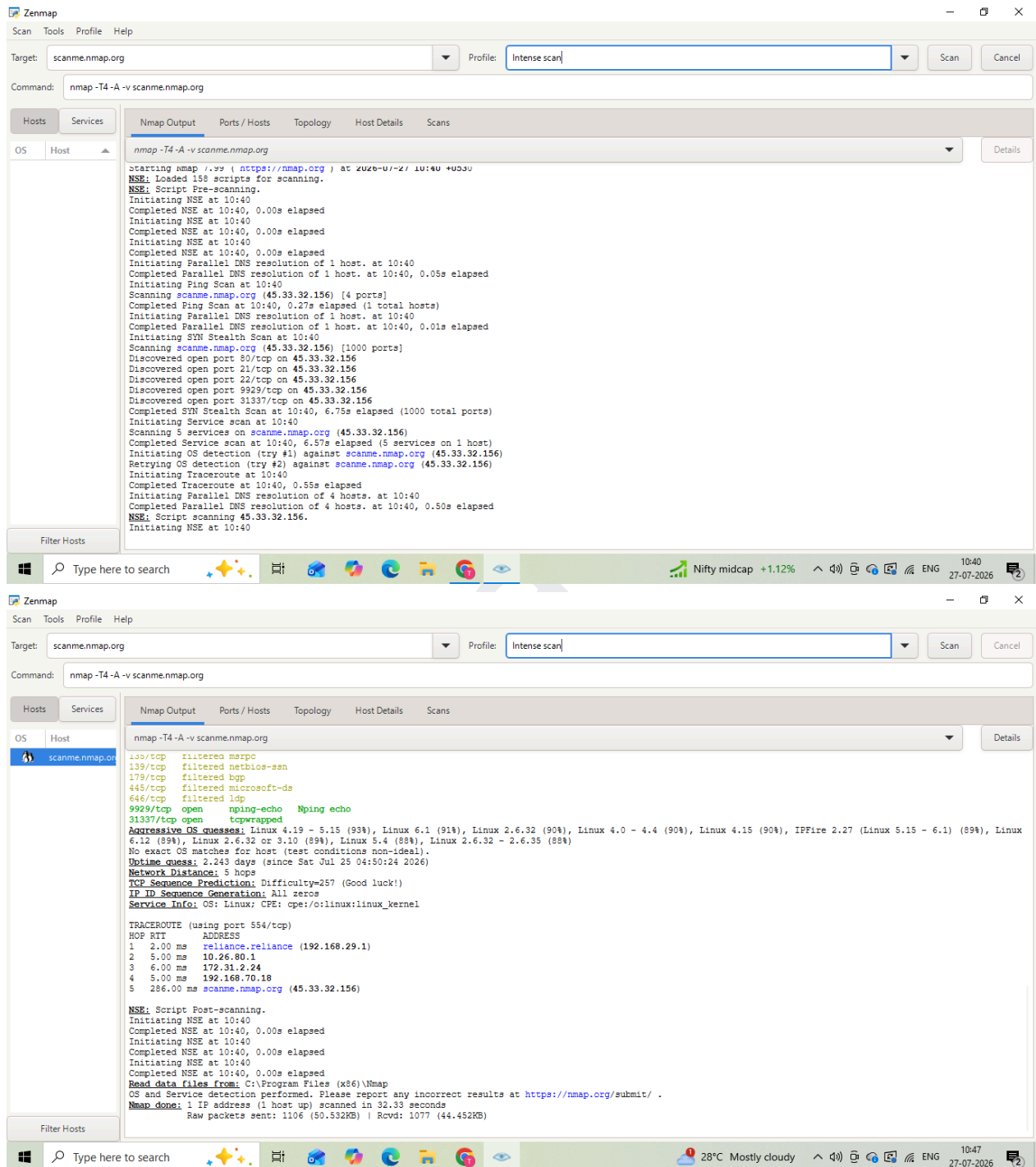
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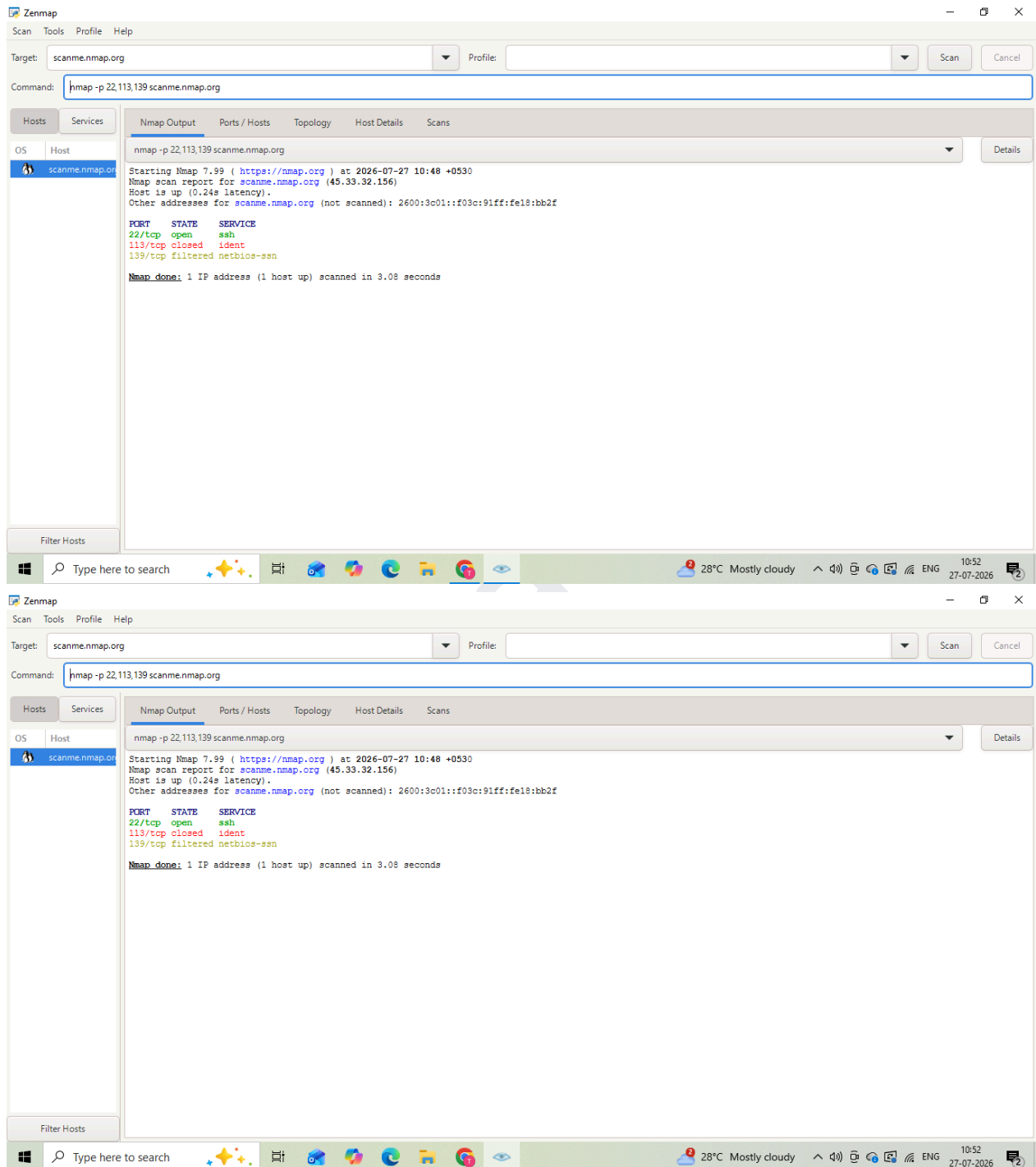
The image displays two screenshots of the Zenmap application interface, showing the results of an Nmap scan performed on the target `scanme.nmap.org` using the `Intense scan` profile.

Top Screenshot: The `Nmap Output` tab is selected. The output shows the scan progress, including the initiation and completion of NSE (Nmap Scripting Engine) scripts, the discovery of open ports (80/tcp, 21/tcp, 22/tcp, 9929/tcp, 31337/tcp), and the completion of the SYN Stealth Scan. The scan was initiated at 10:40 on 2026-07-27.

Bottom Screenshot: The `Nmap Output` tab is selected. The output shows the scan progress, including the initiation and completion of NSE (Nmap Scripting Engine) scripts, the discovery of open ports (135/tcp, 139/tcp, 179/tcp, 445/tcp, 646/tcp, 9929/tcp, 31337/tcp), and the completion of the SYN Stealth Scan. The scan was initiated at 10:40 on 2026-07-27.

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The image displays two screenshots of the Zenmap interface, showing the results of an Nmap scan performed on the target `scanme.nmap.org`.

Top Screenshot:

- Target:** `scanme.nmap.org`
- Profile:** `Intense scan`
- Command:** `nmap -T4 -A -v scanme.nmap.org`
- Hosts:** `scanme.nmap.org`
- Nmap Output:**
 - Completed Parallel DNS resolution of 15 hosts. at 10:58, 0.40s elapsed
 - NSE: Script scanning 45.33.32.156.
 - Initiating NSE at 10:58
 - Completed NSE at 10:58, 7.65s elapsed
 - Initiating NSE at 10:58
 - Completed NSE at 10:58, 2.14s elapsed
 - Initiating NSE at 10:58
 - Completed NSE at 10:58, 0.00s elapsed
 - Nmap scan report for `scanme.nmap.org` (45.33.32.156)
 - Host is up (0.25s latency).
 - Other addresses for `scanme.nmap.org` (not scanned): `2600:3c01::f03c:91ff:fe18:bb2f`
 - Not shown: 989 closed tcp ports (reset)
 - PORT STATE SERVICE VERSION
 - 21/tcp open tcpwrapped
 - 22/tcp open ssh OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
 - 80/tcp open http Apache httpd 2.4.7 ((Ubuntu))
 - 135/tcp filtered msrpc
 - 139/tcp filtered netbios-ssn
 - 179/tcp filtered bgp
 - 445/tcp filtered microsoft-ds
 - 646/tcp filtered ldp
 - 9929/tcp open nping-echo Nping echo
 - 31337/tcp open tcpwrapped
 - Aggressive OS guesses: Linux 4.19 - 5.15 (93%), Linux 6.1 (91%), Linux 2.6.32 (90%), Linux 2.6.32 or 3.10 (90%), Linux 4.0 - 4.4 (90%), Linux 4.15 (90%), IPFire 2.27 (Linux 5.15 - 6.1) (89%), Linux 6.12 (89%), Linux 5.4 (89%), Linux 2.6.32 - 2.6.35 (88%)
 - No exact OS matches for host (test conditions non-ideal).
 - Uptime guess: 2.256 days (since Sat Jul 25 04:50:24 2026)
 - Network Distance: 24 hops
 - TCP Sequence Prediction: Difficulty=255 (Good luck!)
 - IP ID Sequence Generation: All zeros
 - Service Info: OS: Linux; CPE: `oper:/o:linux:linux_kernel`

Bottom Screenshot:

- Target:** `scanme.nmap.org`
- Profile:** `Intense scan`
- Command:** `nmap -T4 -A -v scanme.nmap.org`
- Hosts:** `scanme.nmap.org`
- Nmap Output:**
 - 646/tcp filtered ldp
 - 9929/tcp open nping-echo Nping echo
 - 31337/tcp open tcpwrapped
 - Aggressive OS guesses: Linux 4.19 - 5.15 (93%), Linux 6.1 (91%), Linux 2.6.32 (90%), Linux 2.6.32 or 3.10 (90%), Linux 4.0 - 4.4 (90%), Linux 4.15 (90%), IPFire 2.27 (Linux 5.15 - 6.1) (89%), Linux 6.12 (89%), Linux 5.4 (89%), Linux 2.6.32 - 2.6.35 (88%)
 - No exact OS matches for host (test conditions non-ideal).
 - Uptime guess: 2.256 days (since Sat Jul 25 04:50:24 2026)
 - Network Distance: 24 hops
 - TCP Sequence Prediction: Difficulty=255 (Good luck!)
 - IP ID Sequence Generation: All zeros
 - Service Info: OS: Linux; CPE: `oper:/o:linux:linux_kernel`
 - TRACEROUTE (using port 110/tcp)
 - HOP RTT ADDRESS
 - 1 2.00 ms `reliance.reliance` (192.168.29.1)
 - 2 5.00 ms `10.26.80.1`
 - 3 5.00 ms `172.31.2.24`
 - 4 5.00 ms `192.168.70.18`
 - 5 4.00 ms `172.26.76.165`
 - 6 6.00 ms `172.26.76.130`
 - 7 7.00 ms `192.168.7.244`
 - 8 ... 9
 - 10 28.00 ms `103.198.140.64`
 - 11 236.00 ms `49.45.4.103`
 - 12 227.00 ms `49.45.4.103`
 - 13 ...
 - 14 243.00 ms `ae1.3501.edge7.LosAngeles1.net.lumen.tech` (4.69.219.41)
 - 15 241.00 ms `4.59.59.94`
 - 16 242.00 ms `ae3.r23.lax01.icn.netarch.akamai.com` (23.192.3.87)
 - 17 228.00 ms `ae3.r23.lax01.icn.netarch.akamai.com` (23.192.3.87)
 - 18 244.00 ms `ae17.r02.sjc01.icn.netarch.akamai.com` (23.193.112.211)
 - 19 232.00 ms `ae2.r11.sjc01.icn.netarch.akamai.com` (23.207.232.39)
 - 20 245.00 ms `ae22.gw4-sjc1.netarch.akamai.com` (23.203.158.33)
 - 21 ... 23
 - 24 244.00 ms `scanme.nmap.org` (45.33.32.156)

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The screenshot shows the Zenmap application window. The 'Target' field is set to 'scanme.nmap.org' and the 'Profile' is 'Intense scan'. The command entered is 'nmap -T4 -A -v scanme.nmap.org'. The 'Nmap Output' tab is selected, displaying the following scan results:

```
nmap -T4 -A -v scanme.nmap.org
646/tcp filtered ldap
9929/tcp open  nginx-echo  Nping echo
31337/tcp open  tcpwrapped
Aggressive OS guesses: Linux 4.19 - 5.15 (93%), Linux 6.1 (91%), Linux 2.6.32 (90%), Linux 2.6.32 or 3.10 (90%), Linux 4.0 - 4.4 (90%), Linux 4.15 (90%), IPFire 2.27 (Linux 5.15 - 6.1) (89%), Linux 6.12 (89%), Linux 5.4 (89%), Linux 2.6.32 - 2.6.35 (88%)
No exact OS matches for host (test conditions non-ideal).
Uptime guess: 2.256 days (since Sat Jul 25 04:50:24 2026)
Network Distance: 24 hops
TCP Sequence Prediction: Difficulty=255 (Good luck!)
IP ID Sequence Generation: All zeros
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 110/tcp)
HOP RTT      ADDRESS
1 2.00 ms  reliance.reliance (192.168.29.1)
2 5.00 ms  10.26.80.1
3 5.00 ms  172.31.2.24
4 5.00 ms  192.168.70.18
5 4.00 ms  172.26.76.165
6 6.00 ms  172.26.76.130
7 7.00 ms  192.168.7.244
8 ...
10 28.00 ms 103.198.140.64
11 236.00 ms 49.45.4.103
12 227.00 ms 49.45.4.103
13 ...
14 243.00 ms ae1.3501.edge7.LosAngeles1.net.lumen.tech (4.69.219.41)
15 241.00 ms 4.59.59.94
16 242.00 ms ae3.r23.lax01.icn.netarch.akamai.com (23.192.3.87)
17 228.00 ms ae3.r23.lax01.icn.netarch.akamai.com (23.192.3.87)
18 244.00 ms ae17.r02.sjc01.icn.netarch.akamai.com (23.193.112.211)
19 232.00 ms ae2.r11.sjc01.icn.netarch.akamai.com (23.207.232.39)
20 245.00 ms ae22.gw4-scl1.netarch.akamai.com (23.203.158.53)
21 ...
22 ...
23 ...
24 244.00 ms scanme.nmap.org (45.33.32.156)
```